



LARGER IN MALE BRAIN



Hypothalamus

Certain areas governing male-typical sexual behavior and responses to stress in the hypothalamus are larger in heterosexual males than in females or homosexual males.

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Amygdala

The amygdala, involved in emotional responses, making decisions, and forming emotional memories, is slightly larger in males. However, differences in functions such as responses to negative versus positive emotional stimuli, are more significant.

Brain structures

There are several areas in which quantifiable physical differences have been identified between male and female adult brains. The main regions are shown here. How these differences can affect cognition and psychology are currently the matter of ongoing scientific research.

NONBINARY BRAINS

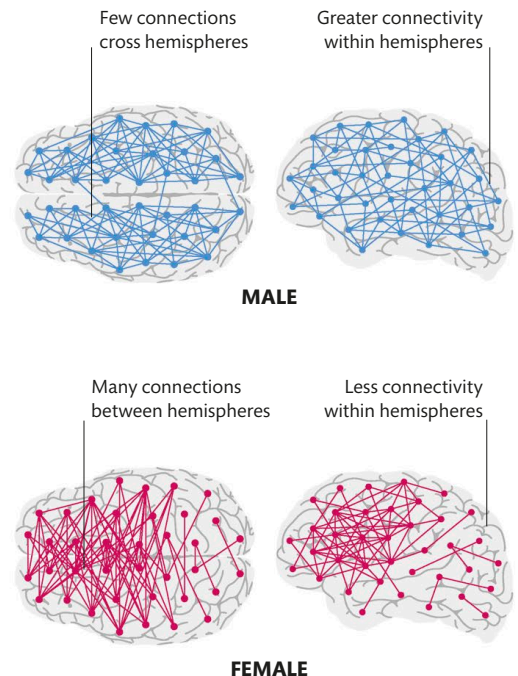
Homosexual and transgender people have been found to have certain distinctive brain structures. For example, some parts of the hypothalamus (see above) differ in homosexual and heterosexual men, and the putamen (involved in learning and regulation of movement) has more gray matter in trans women than in cisgendered men.



NONBINARY
SYMBOL

Differences in function

Male and female brains differ in function as well as structure. Male brains seem to be more “lateralized” (with a greater difference in function between the left and right hemispheres). Males also vary more than females in cognitive ability. These variations are partly due to the structure of the “connectome”—the network of neural connections between parts of the brain (see below). They also result from the action of hormones, and external influences, throughout our life. In particular, our social environment and experiences continually shape our neural pathways, helping us perform male- or female-typical tasks.



The connectome

One study, in which more than 900 brains were imaged, found that male brains have greater connectivity within hemispheres, while female brains have denser connections between hemispheres. The males were found to be better at spatial processing, while the females scored higher on attention and memory for words and faces.